Random Sampling and Sampling Distribution

In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
```

In [2]:
```
population_mean = 50
population_std = 10
population_size = 100000
population = np.random.normal(population_mean, population_std, population_si
```

In [3]:
```
sample_sizes = [30, 50, 100]
num_samples = 1000



sample_means = {}
```

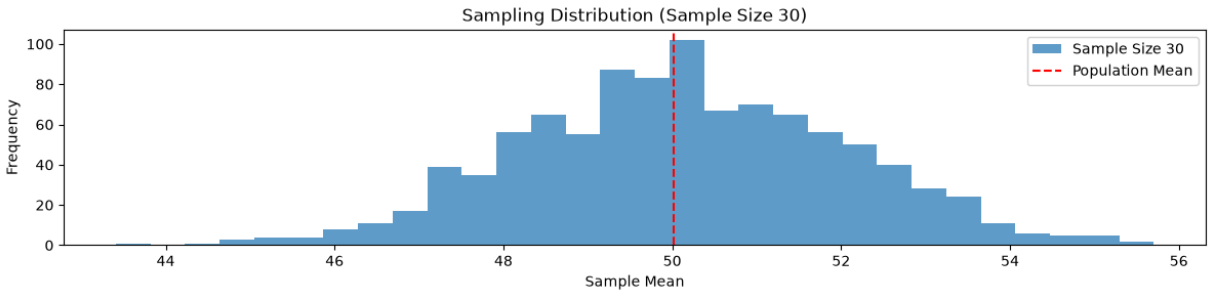
In [4]:
```
for size in sample_sizes:
 sample_means[size] = []
 for _ in range(num_samples):
   sample = np.random.choice(population, size=size, replace=False)
   sample_means[size].append(np.mean(sample))
```

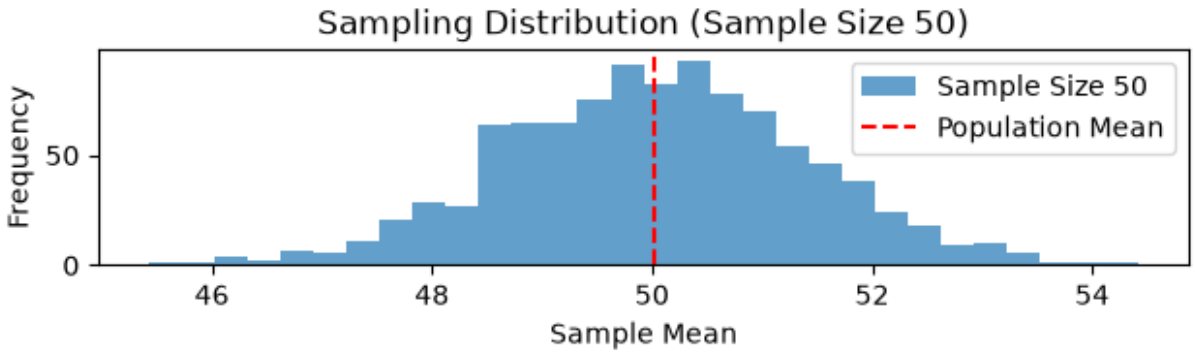
In [6]:
```
plt.figure(figsize=(12, 8))

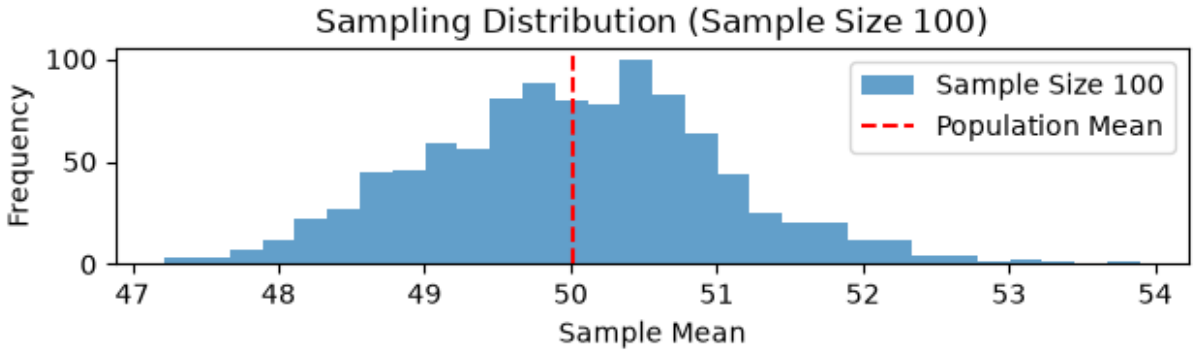
for i, size in enumerate(sample_sizes):
 plt.subplot(len(sample_sizes), 1, i+1)
 plt.hist(sample_means[size], bins=30, alpha=0.7, label=f'Sample Size {size}
 plt.axvline(np.mean(population), color='red', linestyle='dashed', linewidth

 plt.title(f'Sampling Distribution (Sample Size {size})')
 plt.xlabel('Sample Mean')
 plt.ylabel('Frequency')
 plt.legend()

 plt.tight_layout()
 plt.show()
```




Sampling Distribution (Sample Size 50)
Frequency
0
50
46
48
50
52
54
Sample Mean
Sample Size 50
Population Mean


Sampling Distribution (Sample Size 100)
Frequency
0
50
100
47
48
49
50
51
52
53
54
Sample Mean
Sample Size 100
Population Mean

In [ ]: